
📘 User Manual –Temperature and AQI BI Dashboard

🟦 1. Introduction

🔷 Overview

The **BI Temperature and AQI Dashboard** is a Power BI solution designed to analyze:

- Climate and temperature trends
- Air Quality Index (AQI) patterns
- Wildfire activity
- Environmental correlations (GHG, temperature, pollution)

The dashboard integrates multiple datasets and provides **interactive insights for climate analysis across regions**.

🔷 Objective

- Monitor long-term climate patterns
- Identify pollution trends and high-risk regions
- Analyze relationships between environmental factors

🟦 2. Dashboard Structure (Actual Pages)

The dashboard consists of the following report pages:

Page Name	Description
Dashboard Summary	Overall KPIs, trends, and correlations
Temperature	Temperature trends, anomalies, heatwaves

Page Name	Description
Correlations	Relationships between temperature, AQI, and GHG
AQI Atlantic Station Level	AQI insights for Atlantic region
Worst AQI Atlantic	Worst pollution cases in Atlantic
Non-Atlantic AQI Station Level	AQI outside Atlantic
Worst AQI Non-Atlantic	Highest pollution in non-Atlantic regions

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🟦 3. Navigation Instructions

🔹 Page Navigation

- Use report tabs at the bottom
- Use action buttons (if available) to switch between views

🔹 Filters (Slicers)

Common filters available across pages:

- Year
- Month
- Province
- Station
- Pollutant
- Region (Atlantic / Non-Atlantic)

👉 All visuals update dynamically based on slicer selection

🟦 4. Dashboard Components

📊 4.1 KPI Cards

Displayed in multiple pages:

- Average Monthly Temperature
- Avg Station AQI
- Total Fires
- Total Area Burned

👉 Provide quick summary of system state

📈 4.2 Time-Series Visuals

Examples from your dashboard:

- **Max Temp (°C)** line chart
- **Worst Pollutant AQI** trend
- **Temperature anomalies over time**

👉 Used to identify trends and seasonality

📊 4.3 Column / Bar Charts

Used for:

- Trend (°C per decade)
- AQI category distribution
- Days in AQI category

🔄 4.4 Combo Charts

Key visuals include:

- **Area Burned + Total Fires**
- **Heatwave Count + Average Anomaly**

👉 Compare multiple metrics in one view

📌 4.5 Tables & Matrices

Used for:

- Correlation analysis
- Station-level AQI data
- Detailed breakdowns

🟦 5. Key Metrics (Actual from Model)

🌡️ Temperature Metrics

Metric	Description
Avg Monthly Temp	Average monthly temperature
Temp Monthly Anomaly	Difference from baseline
Trend (°C/decade)	Long-term temperature change
Heatwave Count	Number of heatwave events

🌍 AQI Metrics

Metric	Description
Avg Station AQI	Average air quality index
Worst Pollutant AQI	Maximum pollutant level
% Unhealthy Days	Days where AQI > 100
Dominant Pollutant	Most frequent pollutant

🔥 Wildfire Metrics

Metric	Description
Total Fires	Total recorded fires

Metric	Description
Total Area Burned	Land affected
Fire YoY Growth	Year-over-year change
Large Fire Ratio	Proportion of large fires

🔗 Correlation Metrics

- Temperature vs Pollution
- GHG vs Temperature
- Weather vs AQI

👉 Derived using correlation measures in the model

🟦 6. How to Use the Dashboard

✅ Step 1: Select Filters

Choose:

- Year
- Province
- Pollutant

✅ Step 2: Analyze Summary

Start from: 👉 Dashboard Summary page

✅ Step 3: Explore Details

Go to:

- Temperature → climate patterns
- AQI pages → pollution analysis
- Correlation → deeper insights

✅ Step 4: Drill Interaction

- Click visuals to cross-filter
- Hover for tooltips
- Use slicers for segmentation

🟦 7. Interactivity Features

- ✅ Cross-filtering across visuals
- ✅ Drill-down capability
- ✅ Tooltips for detailed values
- ✅ Dynamic slicers

🟦 8. Data Model Overview

The dashboard uses a **star schema model**:

🔹 Fact Tables

- Fact_mClimate
- Fact_AQI
- Fact_Heatwaves
- Fact_Daily_Anomaly

🔹 Dimension Tables

- Model Calendar
- Dim_Stations_All

- Dim_Wildfire
- Dim_temperature_actual_forecast

🟦 9. Limitations

- Data depends on historical sources
- Some metrics are aggregated
- Predictions (forecast) may contain uncertainty
- No real-time data updates

🟦 10. Troubleshooting

Issue	Solution
Dashboard not loading	Refresh page
Blank visuals	Reset filters
Wrong values	Check slicer selections
No data for region	Try different filters

🟦 11. Best Practices

- Start from summary page
- Filter by year before analysis
- Compare multiple regions
- Use correlation page for deeper insights

🟦 12. Conclusion

The BI Temperature dashboard enables:

✅ Climate trend analysis
✅ AQI monitoring
✅ Environmental correlation insights
✅ Data-driven decision-making